Investigating how reward retroactively improves memory for associations and items

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Abstract

Given we cannot remember all our experiences, it seems intuitive that we should prioritize events of consequence, like those that lead to a reward. However, rewards have had inconsistent effects on people's memory for preceding neutral experiences in experiments, raising questions as to whether our minds are equipped to remember adaptively. Here, we used online testing in a registered report to investigate whether rewards tend to retroactively enhance certain types of memories. Consistent with prior research and theories involving how dopamine modulates the hippocampus, we hypothesized that reward would retroactively enhance memory in a delay-dependent manner, emerging only after a period of consolidation. Crucially, we further predicted that this long-delay modulation of memory would be stronger for associative memory than for item recognition—a factor not previously considered despite its clear theoretical grounding. Our planned confirmatory analyses did not reveal any evidence for retroactive effects of reward on either associative or item memory. However, in exploratory analyses examining only individuals who were behaviourally sensitive to the reward manipulation, we found evidence that reward retroactively enhanced associative—but not item—memory in a delay-dependent manner. This work sheds light important open questions about the mechanisms by which reward retroactively enhances memory, elucidating the conditions under which this key aspect of adaptive memory emerges.

Keywords: associative memory, recognition memory, reward, consolidation, dopamine

Public Significance: This study seeks to uncover how rewarding experiences influence memories for the events that precede them. Past work has revealed mixed evidence as to whether rewards can retroactively enhance memory. Here, we aimed to reconcile these mixed findings by testing the theoretically grounded prediction that reward should enhance how we remember associations more strongly than the how we recognize the items comprising them. We found that reward improved memory for preceding associations, but not items. Crucially, this effect emerged only in individuals who were more motivated by the reward itself. These findings help explain why previous studies have produced inconsistent results and reveal the circumstances under which rewards are most likely to shape memory. Overall, our study provides insight into how rewarding experiences shape our memories and guide choices in daily life.

Introduction

Prioritizing useful experiences in our memory is the cornerstone of adaptive behaviour (Richards & Frankland, 2017; Shohamy & Adcock, 2010). How does our memory system determine which of our many experiences should be remembered over time? Intuitively, experiences that allow us to anticipate and re-experience rewards are more important to remember than those that were less consequential, so we can seek out situations that led to a positive outcome in the past. Yet, this prioritization is often more complicated than simply remembering a rewarding experience because many events are seemingly insignificant in the moment but only later become meaningful through their association with a positive outcome (Hull, 1943). In other words, it is adaptive to retroactively enhance memories for neutral experiences that ultimately lead to reward. Although it is well known how such a mechanism might work for automatically conditioned stimuli (i.e., classical conditioning), it is less clear how neutral episodic memories might be retroactively enhanced by reward. How can we enhance memories for these events when their outcomes only later reveal themselves?

Despite the centrality of this problem to understanding adaptive learning and memory (Brzosko et al., 2015; Hull, 1943; Izhikevich, 2007), whether reward does in fact retroactively enhance human memory is unclear due to mixed findings. While some previous studies have found evidence for such an effect (Braun et al., 2018; Murayama & Kitagami, 2014; Patil et al., 2017), others have not (Oyarzún et al., 2016; Sukumaran et al., 2023), raising important questions about the particular circumstances under which it manifests. Identifying the boundary conditions of retroactive, reward-related memory enhancement is crucial to characterizing when different adaptive memory functions operate. Here, we explored a neurobiologically-grounded factor that could reconcile mixed results: namely, distinctions between how we remember associations and recognize the items comprising them. Using a well-powered, registered design

we tested the hypothesis that rewards enhance how we form rich, relational memories of preceding experiences, but are less beneficial for forming cruder memories sufficient to support item recognition.

Several previous studies of human behaviour have elegantly shown that reward can retroactively enhance memory for preceding neutral information (Braun et al., 2018; Murayama & Kitagami, 2014; Patil et al., 2017). In these studies, participants typically view individual items (like pictures of objects) and then have a chance to win a reward in an unrelated game. Using this structure, for example, Murayama and Kitagami (2014) had participants view a single picture of an object and then play one of two games: one that offered monetary rewards and one that did not. The researchers found that people were better at later recognizing items that preceded the reward compared to the no reward game (Murayama & Kitagami, 2014), hereafter referred to as the retroactive reward effect. This finding has been echoed across other similar studies (Braun et al., 2018; Patil et al., 2017). Notably, these studies also show that the retroactive reward effect is delay-dependent, emerging only when memories are tested after a period of consolidation but not immediately (Braun et al., 2018; Murayama & Kitagami, 2014; Patil et al., 2017). This delay-dependence contrasts with studies in which the reward cue precedes (and motivates the learning of) a neutral event (proactive reward), which have found immediate reward-related memory benefits (Gruber et al., 2016; Murty & Adcock, 2014). One potential explanation for the difference between retroactive vs. proactive reward paradigms is that they engage different cognitive mechanisms. Namely, proactive reward paradigms may additionally benefit from enhanced motivational states elicited by reward anticipation, whereas retroactive reward paradigms can isolate the effect of reward because reward (vs. no reward) trials only become apparent after encoding has taken place. These mechanisms can explain the discrepancy in delay-dependency of memory effects: While attentional boosts elicited by reward

motivation should enhance memory immediately, the effect of reward isolated from attention or motivation may take longer to emerge.

Indeed, the striking delay-dependence of the retroactive reward effect was impressively predicted from earlier neurobiological research in rodents. Reward is associated with dopamine release (Montague et al., 2004; Schultz, 1998, 2002), and in turn dopamine strengthens connections between neurons in the hippocampus (Lisman et al., 2011; Lisman & Grace, 2005; Tsetsenis et al., 2021), which are critical for forming memories (Scoville & Milner, 1957). Even recently activated connections are strengthened by dopamine (Frey & Morris, 1997, 1998; Redondo & Morris, 2011), explaining how reward can retroactively enhance memory (de Carvalho Myskiw et al., 2013; Dunsmoor et al., 2022; Moncada & Viola, 2007; Morris, 2006; Salvetti et al., 2014; Wang et al., 2010). But, because it takes time to remodel neural connections, dopamine's effect on both the hippocampus (Bethus et al., 2010; Lisman et al., 2011; O'Carroll et al., 2006; Rossato et al., 2009) and memory performance (Wang et al., 2010) are typically seen only after several hours—mirroring the effect of reward on human memory. Rodent research also shows that dopamine may further retroactively enhance our memory by fostering the replay of hippocampal memories formed prior to either a reward (Ambrose et al., 2016; Singer & Frank, 2009) or dopamine stimulation (McNamara et al., 2014). Together, these effects of dopamine in the hippocampus could mechanistically explain how rewards can reach back in time to enhance our memory for previously unimportant events.

However, this seemingly straightforward story is complicated by other studies of human behaviour showing that rewards do not retroactively enhance memory (Oyarzún et al., 2016; Sukumaran et al., 2023), raising questions about the reliability and boundary conditions of this phenomenon. Why are retroactive reward effects on human memory so variable? While these prior studies vary in many regards (e.g., the semantic relatedness of stimuli), here, we explore a

neurobiologically-grounded possibility. Specifically, past work on this topic has tended to assess a type of memory not necessarily reliant on the hippocampus—item recognition—despite this region's central role in the underlying theories. Specifically, the hippocampus is thought to rapidly bind the disparate aspects of an experience into a memory so that it can later be retrieved with rich contextual details—that is, support associative memory (Davachi, 2006; Eichenbaum, 2000; Mayes et al., 2007). Item memory tends to instead be dependent on other brain regions, such as the perirhinal cortex (Davachi, 2006). Indeed, reflecting these neural differences, past work has shown behavioral distinctions between memories for associations and the single items comprising them (Bisby & Burgess, 2014; Giovanello et al., 2003; Goldfarb et al., 2019; Mayes et al., 2004; Murty et al., 2016). Most relevant to our question, different motivational states engaged in the moments prior to memory formation benefit associative more than item memory (Murty & Adcock, 2017).

While these findings of a dissociation between item and associative memory support the notion that a similar one may also exist in the context of retroactive reward effects, to our knowledge no past work on this topic has probed associative aspects of memory. Instead, all studies have required participants to simply recognize individual stimuli (i.e., assessing item memory; Braun et al., 2018; Murayama & Kitagami, 2014; Oyarzún et al., 2016; Patil et al., 2017; Sukumaran et al., 2023). While item recognition could certainly benefit from the retrieval of contextually rich, hippocampal memories, it importantly does not require them (Brown & Aggleton, 2001; Davachi et al., 2003; Davachi & Wagner, 2002; Eldridge et al., 2000; Wais, 2011; Wan et al., 1999). By only measuring subsequent item recognition, prior research may have underestimated the power of rewards to retroactively enhance hippocampally dependent aspects of memory, and the noise associated with this design decision could drive the inconsistencies seen in the literature.

Here, we explore whether the variability in how rewards retroactively enhance memory

may stem, in part, from the use of specific kinds of item recognition assessments that do not consistently engage the hippocampus. Certain design differences between experiments that have found positive versus null retroactive reward effects hint that this may be the case. For example, in Braun et al. (2018), participants learned items while navigating through a maze, yielding a reliable reward modulation effect; such a spatial memory task could have engaged the hippocampus more than viewing objects on the screen alone (Burgess et al., 2002; Eichenbaum et al., 1999). Further, Patil et al. (2017) found a reliable retroactive reward effect when limiting their analyses to high confidence recognition responses (c.f. Sukumaran et al., 2023), which may indicate retrieval of contextually rich, hippocampal dependent memories despite the item recognition task (Rugg et al., 2012). Consistent with the potential importance of this factor, Oyarzun et al. (2016) collapsed across all trials, regardless of confidence, in their analyses and reported null effects of reward. Another study, however, used recognition memory tests without confidence ratings or a clear link to hippocampal memory and still showed retroactive reward effects (Murayama & Kitagami, 2014). Overall, given this mixed picture emerging from the literature, we need a systematic test of the prediction that reward retroactively enhances hippocampally dependent memories. As such, we directly compared the retroactive reward effect on item recognition and associative memory to determine if retroactive enhancements are significantly more robust for associative aspects of memory, which have been closely linked to hippocampal function (Davachi, 2006; Eichenbaum, 2000).

In this study, we investigated the retroactive effect of reward on item and associative memory using a well-powered, registered, within-subjects design. We modified a task from Murayama and Kitagami (2014) to incorporate features from Braun et al. (2018). Unlike Murayama and Kitagami (2014), who exposed participants to a single object, we exposed participants to short lists of object-object associations, which were either followed by an

unrelated reward ($1) or no reward ($0) game. Our inclusion of object pairs instead of single objects allowed us to directly compare how reward retroactively enhances associative and item memory. Specifically, following the incidental memory formation phase, we immediately tested participants' memory for (1) half of the objects (item memory) and (2) the corresponding object-object associations (associative memory). Participants were tested again one day later, at which point we tested their memory for the other half of the objects and object-object associations. Moreover, including six encoding trials per reward game rather than one additionally allowed us to test if the retroactive effect of reward depends on its temporal distance from the encoding event (Braun et al., 2018). We also included a filler task after each encoding block to ensure that reward effects did not spill over into the subsequent set of images, as in Murayama and Kitagami (2014).

We investigated three pre-registered, *a priori* hypotheses. First, we hypothesized a retroactive reward effect on memory for object-object associations, but only when tested at a delay. Specifically, we hypothesized that the effect of reward on associative memory would depend on delay (Hypothesis 1a, see Table 1), such that reward would not influence associative memory in the immediate test (Hypothesis 1b, see Table 1), but would reliably enhance associative memory in the delayed test (Hypothesis 1c, see Table 1).

Next, we investigated whether there is a delay-dependent effect of reward on item recognition (Hypothesis 2a, see Table 1). Consistent with past work (Braun et al., 2018; Murayama & Kitagami, 2014; Patil et al., 2017), we hypothesized that subsequent rewards would not influence item recognition on the immediate test (Hypothesis 2b, see Table 1). We did not have strong predictions about reward modulation at the long delay (Hypothesis 2c, see Table 1). On the one hand, Murayama & Kitagami (2014) found that delayed item recognition was retroactively enhanced by reward; as such, the same might be expected here. On the other

hand from a theoretical perspective we did not expect a strong role for the hippocampus in item memory, nor empirically did we see any evidence for this effect in our pilot data (see Supplementary Information).

Finally, we directly compared retroactive reward modulation across memory test types (item vs. associative) using normalized scores. We hypothesized that the retroactive reward effect would be larger for associative than item memory when tested at a delay (Hypothesis 3, see Table 1). For all of our hypotheses, we interpreted significant positive effects from null hypothesis significance testing in the predicted or opposite direction, and in the case of null results performed Bayesian equivalence testing using the ROPE procedure (Kruschke, 2018) to assess evidence for null hypotheses as detailed in **Table 1**.

Table 1 | Design Table

	Question	Hypothesis	Analysis Plan
Associative Memory	**1a** Is there a delay-dependent effect of reward on associative memory?	There will be a significant interaction between reward and delay, such that the effect of reward on associative memory will be larger for the delayed vs. immediate test.	We will conduct a frequentist linear mixed effects model predicting average associative memory accuracy as a function of reward, delay, and their interaction, with random intercepts grouped by participant. If there is no significant coefficient for reward or the interaction between reward and delay, we will conduct a Bayesian mixed effects model predicting average associative memory accuracy as a function of reward, delay, and their interaction, with random intercepts grouped by participant. We will then determine evidence for the null* (see *Methods*).
	1b Is there a retroactive effect of reward on associative memory in the immediate test?	Associative memory will show no meaningful differences between the reward and no reward conditions in the immediate test.	We will obtain the contrast between reward and no reward on associative memory for the immediate test using the frequentist linear mixed effects model described in Hypothesis 1a. If there is no significant coefficient for reward, we will obtain the reward vs. no reward contrast for the immediate test from the Bayesian mixed effects model described in 1a. We will then determine evidence for the null* (see *Methods*).
	1c Is there an effect of reward on associative memory in the delayed test?	Associative memory accuracy will be significantly higher in the reward compared to the no reward condition in the delayed test.	We will obtain the contrast between reward and no reward on associative memory for the delayed test using the frequentist linear mixed effects model described in Hypothesis 1a. If there is no significant coefficient for reward, we will obtain the reward vs. no reward contrast for the delayed test from the Bayesian mixed effects model described in 1a. We will then determine evidence for the null* (see *Methods*).
Item Memory	**2a** Is there a delay-dependent effect of reward on item recognition?	Given mixed findings in the literature, we are agnostic as to whether there will be an interaction between reward	We will conduct a frequentist linear mixed effects model predicting item recognition (defined as corrected recognition: hit rate-false alarm rate) as a function of reward, delay, and their interaction, with random intercepts grouped by participant.

		and delay on item recognition.	If there is no significant coefficient for reward or the interaction between reward and delay, we will conduct a Bayesian mixed effects model predicting average associative memory accuracy as a function of reward, delay, and their interaction, with random intercepts grouped by participant. We will then determine evidence for the null* (see *Methods*).
	2b Is there a retroactive effect of reward on item recognition in the immediate test?	Item recognition will show no meaningful differences in the reward and no reward conditions in the immediate test.	We will obtain the contrast between reward and no reward on item recognition (corrected recognition: hit rate-false alarm rate) for the immediate test using the frequentist linear mixed effects model described in Hypothesis 2a. If there is no significant coefficient for reward, we will obtain the reward vs. no reward contrast for the immediate test from the Bayesian mixed effects model described in 2a. We will then determine evidence for the null* (see *Methods*).
	2c Is there a retroactive effect of reward on item recognition in the delayed test?	Given mixed findings in the literature, we are agnostic as to whether item recognition will be modulated by retroactive reward in the delayed test.	We will obtain the contrast between reward and no reward on item recognition (corrected recognition: hit rate-false alarm rate) for the delayed test using the frequentist linear mixed effects model described in Hypothesis 2a. If there is no significant coefficient for reward, we will obtain the reward vs. no reward contrast 2a for the delayed test from the Bayesian mixed effects model described in. We will then determine evidence for the null* (see *Methods*).

Comparing Associative to Item Memory	**3** Is the effect of reward greater for associative than item memory in the delayed test?	There will be a significant interaction between memory test and reward, such that the effect of reward on normalized z scores is larger for associative than item memory.	We will z score average accuracy in the associative memory test and corrected recognition (hit rate-false alarm rate), irrespective of reward and delay. We will then conduct a frequentist linear mixed effect model predicting z score as a function of the interaction between reward (reward vs. no reward) and memory test (item vs. associative test) in the delayed test, with random intercepts grouped by participant. If there is no significant coefficient for the interaction between reward and memory test, we will also conduct a Bayesian mixed effects model predicting z score as a function of the interaction between reward (reward vs. no reward) and memory test (item vs. associative test) in the delayed test, with random intercepts grouped by participant. We will then determine evidence for the null* (see *Methods*).

*evidence for the null is calculated as the high density interval of the Bayesian model that falls within the region of practical equivalence, defined as 0 ± 0.17 times the standard deviation of the dependent variable (see *Methods)*

Methods

Ethics Information.

All procedures were in accordance with a study protocol approved by the Research Ethics Board at the University of Toronto. All participants provided informed consent and were compensated $8 CAD per hour for participating. They also had the opportunity to earn an additional monetary bonus of up to $8 depending on their performance in the reward section of the task (see *Design*).

Sampling Plan.

Participants.

Participants were recruited through CloudResearch, an online participant management tool associated with Mechanical Turk (Litman et al., 2017). To be eligible to participate, they

had to reside in the United States, report that they were between the ages of 18 and 30, report that they speak English well or very well (see Supplementary Appendix 1), and have normal or corrected to normal vision. Participants were asked to provide the following demographic information: gender, sex assigned at birth, age, years of education, race, ethnicity, country of birth, country of current residence, postal code, vision/colourblindness, psychiatric diagnoses, socioeconomic status (household income, monthly expenses, subjective social standing). See Supplementary Information (Supplementary Appendix 1) for the full text and response options of our demographics questionnaire. A participant's data was also excluded from analyses if they failed to complete the second session of the experiment one calendar day after the first session, if they stopped partway through a session, or if their data were not properly recorded or transferred to Inquisit.

Participants were also excluded based on task compliance and performance on the memory tests using a multistep approach. First, we excluded participants who were not compliant (a) in the clock game, defined as less than 75% success rate across both reward and no reward conditions, and/or (b) in the flanker task, defined as less than 50% response rate. This ensured that participants in the final sample were engaged in the task. Within this sample of compliant participants, we conducted analyses on memory performance for each of the four memory tests (item recognition and associative memory in the immediate and delayed sessions), collapsed across reward. We calculated the proportion of participants whose performance was at floor or ceiling (e.g., to determine whether any design changes are required, see *Quality Checks*). Next, we excluded participants if they did not perform numerically above chance (corrected recognition [hit rate-false alarm rate] of 0 in the item recognition test, 33% in the associative memory test) in one or more of the four memory tests. Exclusion-related memory analyses were performed across all trials, irrespective of clock game performance. Data needing to be excluded

after collection were replaced by recruiting new participants until we reach our target sample size (see below).

Power Analyses.

The effect size of retroactive reward modulation at a long delay is uncertain, with reports in the literature ranging from $d = 0.12$ (Sukumaran et al., 2023) to $d = 0.54$ (Murayama & Kitagami, 2014) for item recognition, and there is no published data from associative memory tests. Given this uncertainty, we employed a sequential analysis procedure (Lakens, 2014) with predefined checkpoints, including the lower and upper bounds for our sample size. Conservatively, we selected n = 90 participants who met all inclusion criteria (including performance requirements) as our *minimum* sample size, which provided 95% power to detect an effect size of $d = 0.39$ and 99.9% power to detect the effect size reported by Murayama and Kitagami ($d = 0.54$; Murayama & Kitagami, 2014). We had an additional checkpoint at n = 165. Due to resource and time constraints, we selected n = 240 as our maximum sample size. With this maximum sample size, we had 95% power to detect a small effect size of d = 0.24 or higher using an adjusted alpha of p = 0.044 (see below). This small effect size is based on effect sizes reported in prior work revealing positive retroactive reward effects and in our pilot analyses (see Supplementary Information). Thus, n = 240 would be sufficient to detect an effect using frequentist tests for each of our positive, confirmatory hypotheses (see Table 1). See Supplementary Information for power simulations of null effects, which indicated that we were well-powered to detect simple null effects. Note that our sample size determination did not require 95% power to detect null effects using Bayesian tests because this prohibited the assessment of interactions.

We used the GroupSeq package in R (version 1.3.5; Pahl, 2022) to control the type I error rate at 0.05 by adjusting the alpha boundary at each checkpoint (Checkpoint 1: n = 90, alpha

boundary = 0.0031; Checkpoint 2: n = 165, alpha boundary = 0.0183; Checkpoint 3: n = 240, alpha boundary = 0.044). At each checkpoint, we assessed evidence for the null for all hypotheses that did not reach significance using the Bayesian Region of Practical Equivalence (ROPE; Kruschke, 2018) procedure. First, we defined the ROPE for each analysis as zero ± 0.17 times the standard deviation of our dependent variable for each analysis (boundaries determined to achieve 95% power for simple effects through simulations; see *Supplement* for simulation results). Next, we ran Bayesian models and determined whether 95% of the Highest Density Interval (HDI) fell inside the ROPE (Kruschke, 2018). At a given checkpoint, if all hypotheses had informative answers—either reaching the alpha boundary or yielding sufficient evidence for the null—we planned to stop data collection. Otherwise, we planned to continue data collection until we reached our maximum sample size.

Design.

Materials.

Stimuli consisted of 288 color images of emotionally neutral objects (e.g., lamp, pencil, mug) collected from the BOSS stimulus set (Brodeur et al., 2010). Objects were grouped into pairs chosen to have low semantic and visual similarity to each other. Semantic similarity between objects in a pair were assessed using WordNet::Similarity (Princeton University, 2010), an online lexical database assessing semantic relatedness between words. We used a similarity score of 0.2 as our cut off for semantic relatedness within our pairs. We chose to minimize semantic relations between word pairs because semantic relations enhance people's ability to bind items in memory (Epstein et al., 1975) and, thus, could be an unintended source of noise in our attempt to estimate the retroactive impact of reward on associative memory. Visual similarity was assessed by two independent raters, who were told to give binary responses about the colour and shape similarity of object pairs and then regroup pairs if both raters agreed that objects were

visually similar. Ninety-six object pairs were presented during encoding, and the additional 48 object pairs (96 objects) were reserved and used as lures in the item recognition test (see *Procedure)*. Object pairs were counterbalanced across participants such that each was equally likely to appear during encoding or as lures during the item recognition test, in the rewarded versus not rewarded condition, and in the immediate versus long delay condition.

Procedure.

We conducted a 2 (reward, no reward) x 2 (immediate, long delay), within-subjects experiment adapted from the procedure used by Murayama and Kitagami (2014). The experiment was completed across two sessions (**Figure 1A**). The first session of the experiment consisted of three phases: an encoding phase, an item recognition test, and an associative memory test. Our choices to use object pictures as stimuli and make the encoding task incidental were made in alignment with past work (Murayama & Kitagami, 2014; Patil et al., 2017; Braun et al., 2018). Participants then completed the second session one day later, which consisted of another item recognition and associative memory test. The experiment was programmed in and hosted through Inquisit 5.0 (Millisecond Software). Participants performed the task on their personal computers and were not eligible to participate with a phone or tablet. Data collection and analysis was not performed blind to the conditions of the experiments, but participants were not aware of the experimental manipulations or hypotheses and did not interact with the experimenter during online data collection.

Encoding Phase. The incidental encoding phase consisted of a series of 16 blocks. Each block contained a list of six object pairs followed by a reward modulation event and then six flanker trials. Each block began with the presentation of pairs of objects on a computer screen, with each pair visible for 4 seconds with a 500 millisecond ISI in between object presentations (hereafter referred to as encoding trials; **Figure 1B**). To encourage incidental

memory encoding, participants were instructed to imagine the two depicted objects interacting (no response required) but were not told that their memory for these objects would be tested later. For example, if shown a pencil and a mug, they were told that they could imagine putting the pencil inside the mug. Participants viewed six object-object associations in one block, for a total of 96 object-object associations across the entire experiment. We increased the incidental encoding phase from a single object in Murayama and Kitagami (2014) to a block of six object-object pairs to test the time-course of retroactive reward as a function of distance to the reward (see *Exploratory Analyses*). All objects from a given block were tested in the same delay condition (either in the immediate test or the delayed test, see below).

During the reward modulation event (**Figure 1B**), participants played the "clock game" which was associated with a monetary reward half of the time. First, participants saw a 1 second cue that said "reward clock" on reward trials or "no reward clock" on no reward trials. They then saw a fixation cross for a variable duration of 1.5-2 seconds. Participants then saw a circle appear on the screen and, on both reward and no reward trials, they had to press the spacebar within 500ms. The circle was a different colour (green or blue) to signify reward versus no reward trials, with the colour-condition mapping counterbalanced across participants. On reward trials, there could be one of two outcomes: if participants pressed the spacebar fast enough, they were rewarded with $1, and saw a message that said "Congratulations! You won $1" for 2 seconds. If they were too slow on reward trials, they failed to get a reward and saw a message that said "Too slow! The next trial will begin shortly" for 2 seconds. In comparison, on no reward trials, regardless of how quickly participants pressed the spacebar, they did not get a reward and saw a message that said "The next trial will begin shortly" for 2 seconds. Half of the blocks were from the reward condition and half from the no reward condition. The order of reward and no reward blocks were pseudorandomized across participants to ensure that (1) rewards were equally distributed between the first and second

half of the experiment and (2) participants received no more than two reward blocks in a row. Note that we modified the reward game from Murayama & Kitagami (2014) to include a clock animation and a variable anticipation window prior to the speeded response task, similar to the Monetary Incentive Delay task which has been shown to activate dopaminergic midbrain regions (Knutson et al., 2001).

Following the reward event, participants completed six trials of a flanker task (**Figure 1B**) to reduce the carryover of the reward modulation to memories formed in the next block. Participants saw three arrows on the screen and were told to indicate the direction that the middle arrow was pointing, with the outer arrows randomly selected to point in the same or different direction as the middle arrow. The direction of the middle arrow pointed left and right each half of the time. They had 2 seconds to make their response with a 500 millisecond ISI between trials. If participants failed to respond they automatically progressed to the next trial. In total, the filler task was 15 seconds long. This filler block duration was longer than in Murayama and Kitagami (2014) to further reduce the chance of reward carryover.

Each block took approximately 50 seconds to complete, and the entire encoding phase took approximately 13.5 minutes. Participants had the opportunity to take one self-timed break in the middle of the encoding phase.

Immediate Memory Tests. Immediately following the encoding phase, participants completed an item recognition test (**Figure 1C**) for half of the individual objects presented in the encoding phase. They saw 96 old objects (48 objects from the reward condition and 48 objects from the no reward condition) and 48 lures. Each item was either preceded by an item that had 1) the same reward outcome and was experienced in the same encoding block, 2) the same reward outcome but was experienced in a different block, or 3) the different reward outcome and was experienced in a different block. This enabled us to test whether reward leads to recognition

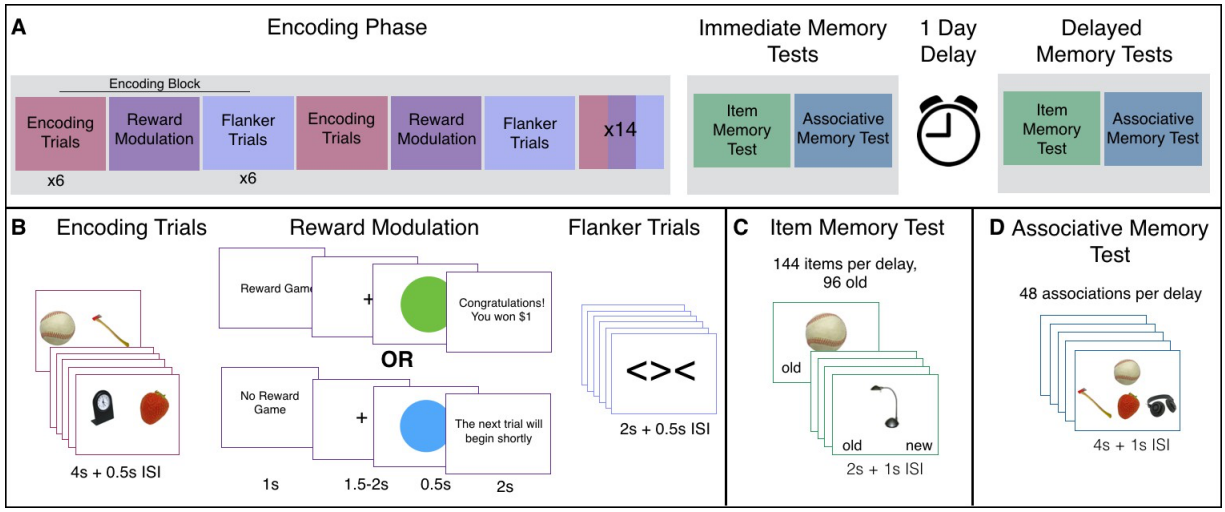
priming for items experienced in the same block (see *Exploratory Analyses*). They were instructed to respond "old" if they believed they saw the object during the encoding session or "new" if they believed that the object was not previously presented. They additionally had an opportunity to rate their confidence alongside their old or new judgements. Therefore, there was four response options: "sure old", "maybe old", "maybe new" or "sure new". Participants had 2 seconds to respond to each item, immediately after which a 1 second ISI began, regardless of when the participants responded. The item recognition test took approximately 6 minutes to complete.

Following the item recognition test, participants completed a 3 alternative forced choice associative memory test (**Figure 1D**) for the object-object associations presented during the encoding phase. The associative memory test always happened after the item recognition test because while the associative memory test re-exposed participants to old objects and therefore may have facilitated their memory, the recognition test did not re-expose participants to associations. In the associative memory test, participants were shown one object at the top of the screen and asked to indicate which of the three objects shown on the bottom was associated with it. Participants were tested on 48 associations, comprising the same objects tested in the item recognition test (24 associations from the reward condition and 24 from the no reward condition). Additionally, the lures in the 3 alternative forced choice associative memory test included one object from a different association encoded in the same block, and one encoded in a different block. All lures were from the same reward and delay condition as the correct answer on a given trial (e.g., if the correct answer on a trial was from the reward, immediate condition, both lures were also from the reward, immediate condition), and therefore were tested in the preceding recognition test. Participants had 4 seconds to respond, immediately after which a 1 second ISI began, regardless of when the participants responded. The associative memory test

took approximately 4 minutes to complete.

Long Delay Memory Tests. Participants were invited to participate in the second session one calendar day after completing session one, with a minimum delay of 12 hours between sessions. They were prohibited from participating in session two if the delay was more than one calendar day. In the second session, participants then completed item recognition and associative memory tests in the same format as described above. In the item recognition test, they were tested on the 96 previously untested objects from encoding (48 objects from the reward condition and 48 objects from the no reward condition) along with 48 novel lures. In the associative memory test, they were tested on the previously untested 48 associations (24 associations from the reward condition and 24 associations from the no reward condition) from encoding. Finally, participants completed a short post-task questionnaire in which they were asked about their strategies during the experiment. To prevent attrition, they were paid their monetary bonus of up to $8, corresponding to their performance on reward trials during incidental encoding in the first session, upon completion of the long delay memory tests.

Figure 1 | Task Schematic



(**A**) Task Overview. In the first session, participants completed the encoding phase and the

immediate memory tests. The encoding phase consisted of encoding trials, a reward modulation event, and flanker trials that were completed sequentially in 16 blocks. Participants then completed item recognition and associative memory tests. Participants returned one day later for a second session in which they completed additional item and associative memory tests. (**B**) Schematic of the Encoding Phase. In each encoding trial, participants encoded an object-object association. Participants then saw a circle that was animated to look like a clock on the screen for 3 seconds (reward modulation event). They were told to press the space bar within 500ms of seeing the circle. On reward trials, if they pressed the space bar quickly enough, they were given $1. On no reward trials, they did not earn any money. Participants then completed flanker trials in which they saw three arrows on the screen and were told to indicate which direction the middle arrow was pointing. The flanker trials acted as a reward washout period before beginning the next block of encoding trials. (**C**) Item Recognition Test. Participants completed an item recognition test for the individual objects learned during the encoding phase. Participants saw an object on the screen and had to indicate whether it was old or new. Half of the objects were tested immediately, and the other half were tested after the delay. (**D**) Associative Memory Test. Participants were then tested on the object-object associations learned during the encoding phase. They saw an object on the top of the screen and had to indicate which of the three objects on the bottom it had been paired with during encoding. Half of the associations were tested immediately, and the other half of the associations were tested after the delay.

Quality Checks.

We performed quality checks on the data to ensure the absence of floor or ceiling effects. This analysis was performed after compliance-related exclusion but before memory- related exclusions (see Participants). We reported the proportion of participants with perfect performance on either memory test (hit rate of 1 and false alarm rate of 0 on the recognition test, 100% accuracy on the associative memory test) and with below chance performance (corrected recognition <0, <33% accuracy on associative memory test). We monitored performance of the first 40 participants. If more than 10% of participants were at floor or ceiling, we planned to stop data collection and adjust the task to make it easier or more difficult, depending on whether participants tended to be at floor or ceiling (e.g., we will increase or decrease the allotted time to respond on the memory tests). However, given the lack of floor of ceiling effects in our pilot data (see below), we reasoned this was unlikely to be necessary. To validate our reward manipulation, we reported the speeding and accuracy differences between reward and no reward trials on the clock game. If there were no significant speeding or accuracy differences between reward conditions (assessed using null hypothesis significance testing) in the first 40 participants, we planned to increase the reward to $1.50. Again, given the robust effect of reward on clock game behavior in our pilot data (see below), we anticipated this was unlikely to be necessary.

Analysis Plan.

We conducted all analyses using a combination of frequentist and Bayesian models in the R programming language. Specifically, we ran frequentist models, consistent with the analysis framework employed in past work (Murayama & Kitagami, 2014; Patil et al., 2017; Braun et al., 2018), and conducted follow up Bayesian analyses for all non-significant effects to evaluate evidence for the null hypothesis, as was done in other registered reports (Berens et al., 2020).

For all analyses related to our hypothesis tests, we restricted our consideration of reward blocks to only those in which the participant successfully obtained the reward. Missed responses on the memory tests were coded as incorrect. To conduct our primary frequentist analyses, we used linear mixed effects models (LMMs, lmer function in the *lme4* package, version 1.1.26; Bates et al., 2015). Because our analyses modeled summary statistics (i.e., corrected recognition [hit rate-false alarm rate]), all models included random intercepts, but not random slopes, grouped by participant. Confidence intervals for models were calculated with the confint function in R (*lme4 package,* version 1.1.26; Bates et al., 2015). If the models failed to converge, we used bobyqa optimization and increased the number of iterations.

For frequentist models did not reveal significant results, we planned to use Bayesian generalized non-linear multivariate multilevel models (*brms* package, version 2.15.0; Bürkner, 2017) to determine evidence for the null hypothesis for each of our *a priori* hypotheses. Bayesian models were fit with the default weakly informative priors, with a student's t distribution centered at 0 with 1 degree of freedom and a scale parameter of .25. Using the *bayestestR* package (version 0.11.5), we then determined the percentage of the HDI that fell within the ROPE (0 ± 0.17; see Supplementary Information for simulations). If 95% of the HDI for the Bayesian model parameters of interest fell inside the ROPE, this would indicate evidence in favour of the null hypothesis. See Supplementary Information for all frequentist and Bayesian model specifications.

Associative Memory. We assessed whether reward influenced associative memory in the immediate and delayed tests (H1a, H1b, and H1c). We first calculated average accuracy on the associative memory test for each participant in each reward and delay condition. This resulted in a single vector of associative memory scores with each participant contributing four numbers—one for each combination of reward and delay condition. We opted to model average

performance rather than trial-wise accuracy to keep this model as similar as possible to our item recognition model (see below). Next, we ran a LMM with average accuracy as the dependent variable and reward (-0.5 = no reward, 0.5 = reward), delay (-0.5 = immediate, 0.5 = delayed), and their interaction as the independent variables. In these models, a significant beta for reward would suggest that reward (vs. no reward) retroactively enhances or impairs associative memory. A significant beta for delay would suggest that associative memory differs between the immediate and delayed tests. A significant beta for the interaction between reward and delay would suggest that the effect of reward on associative memory depends on whether memory is tested immediately or after a delay (H1a). If there was neither a significant effect of reward, nor an interaction between reward and delay, we planned to assess evidence for the null using a Bayesian model with the same structure and the ROPE procedure, as described above.

We also ran planned follow up contrasts on the LMM described above using the emmeans package in R (Lenth et al., 2024). Specifically, we investigated the contrasts between reward and no reward conditions separately for the immediate and delayed tests. In these contrasts, a significant effect for reward would suggest that reward retroactively enhances (or impairs) associative memory in either the immediate test (H1b) or the delayed test (H1c). If there was not a significant effect of reward in the immediate or delayed test, we planned to assess evidence for the null by obtaining the reward vs. no reward contrast from the Bayesian model described above, and then use the ROPE procedure described above.

Item Recognition. Next, we calculated corrected recognition (hit rate – false alarm rate) for each participant in each reward and delay condition. This resulted in a single vector of item recognition scores with each participant contributing four numbers—one for each combination of reward and delay condition. For our confirmatory item recognition analyses, we collapsed across confidence ratings to calculate hit rates (“sure old” or “maybe old” response for old items)

and false alarm rates ("sure old" or "maybe old" response for new items), regardless of confidence. To determine whether reward retroactively enhanced item memory recognition (H2a, H2b, and H2c), we ran a LMM with corrected recognition as the dependent variable and reward (-0.5 = no reward, 0.5 = reward), delay (-0.5 = immediate, 0.5 = delayed), and their interaction as the independent variables. In these models, a significant beta for reward would suggest that reward (vs. no reward) retroactively enhances or impairs item recognition. A significant beta for delay would suggest that item recognition differs between the immediate and delayed tests. A significant beta for the interaction between reward and delay would suggest that the effect of reward on item recognition depends on whether memory is tested immediately or after a delay (H2a). If there was neither a significant effect of reward, nor an interaction between reward and delay, we planned to assess evidence for the null using a Bayesian model with the same structure and the ROPE procedure, as described above.

We again ran planned follow up contrasts on the LMM described above using the emmeans package in R (Lenth et al., 2024). Specifically, we investigated the contrasts between reward and no reward conditions separately in the immediate and delayed tests. In these contrasts, a significant effect for reward would suggest that reward retroactively enhances (or impairs) item memory in either the immediate test (H2b) or the delayed test (H2c). If there was not a significant effect of reward in the immediate or delayed test, we planned to assess evidence for the null by obtaining the reward vs. no reward contrast from the Bayesian model described above, and then use the ROPE procedure described above.

Comparing Item Recognition and Associative Memory. To assess our final a priori hypothesis (H3) that the retroactive effect of reward would be stronger for associative memory as compared to item recognition at the long delay, we put both dependent variables on the same scale. Specifically, for each memory test (item memory and associative memory), we averaged

each participant's performance, separately for the reward and no reward conditions in the immediate and delayed tests. This created a vector of scores for each memory test, with each participant contributing four scores: a reward and a no reward score in the immediate and delayed tests. We then z scored these vectors separately for each memory test, irrespective of delay and reward. We then ran an LMM with the normalized score as the dependent variable, and memory test (-0.5 = item recognition, 0.5 = associative memory), reward (-0.5 = no reward, 0.5 = reward), and their interaction as the independent variables. Because the effect of reward should be strongest after a delay, our confirmatory analyses only included the data from the delayed session. A significant interaction would support our hypothesis (H3) if the underlying pattern derives from H1a and H2a also being supported. A significant interaction in the absence of support for H1a and H2a would be inconsistent with our predictions, as would a non-significant interaction. If there was not a significant interaction between reward and memory test, we assessed evidence for the null using a Bayesian model with the same structure and the ROPE procedure, as described above.

If we did not find a significant effect of reward on either recognition or associative memory with our final sample size, we planned to try running analyses on a subset of participants who performed statistically above chance on all trials (i.e., exhibit both above-chance item recognition, determined by bootstrapping the null for each participant; as well as above-chance associative memory performance of over 47.91% accuracy, determined using a binomial test). However, this did not influence our sampling plan, which was based on memory performance being numerically above chance (see *Sampling Plan)*.

Exploratory Analyses. We conducted the following registered exploratory analyses (described in more detail below): (1) assessed high-confidence item memory by restricting analyses to trials that received high confident judgements; (2) assessed associative memory by

restricting analyses to trials for which both constituent items were correctly recognized; (3) assessed whether greater reward modulation of associative vs. item memory was delay-dependent; (4) asked whether proximity to reward influenced effects by both (a) characterizing the continuous effect of distance to reward and (b) restricting analysis to trials encoded immediately preceding reward; and (5) tested whether hit rate was higher for trials preceded by items that were encoded in the same block.

First, we investigated whether item memory (corrected recognition) was influenced by reward, delay, and their interaction for high confidence responses only (the proportion of old trials [hits] and new trials [false alarms] to which the participant responded "sure old"), consistent with the analysis used in Patil et al. (2017). This analysis was restricted to participants who had at least 5 high confidence responses to each trial type (Reward Old, No Reward Old, and New) per test (immediate and delayed). We also ran planned contrasts to investigate whether high confidence item recognition was influenced by reward in the immediate or the delayed session, separately. Second, we more directly targeted the associative aspect of memory by conditioning the inclusion of trials on correct recognition of both items in the associative pair. We investigated conditional associative memory as a function of reward, delay, and their interaction, and conducted follow up contrasts to investigate conditional associative memory as a function of reward in the immediate and delayed session, separately.

Third, we conducted an exploratory analysis to test whether greater reward modulation of item vs. associative memory was delay-dependent. We conducted a LMM with normalized score as the dependent variable and reward (-0.5 = no reward, 0.5 = reward), delay (-0.5 = immediate, 0.5 = delayed), memory test (-0.5 = item recognition, 0.5 = associative memory), and their interactions as the independent variables. We hypothesized that there would be a significant three-way interaction, such that the stronger effect of reward (vs. no reward) on associative

memory (vs. item recognition) would be greater for the delayed (vs. immediate) session. Other significant patterns would be inconsistent with our hypothesis.

Fourth, we also conducted planned, exploratory analyses to test the time-course of retroactive reward on item and associative memory. We conducted LMMs with item and associative memory at the long delay as the dependent variables, and reward (-0.5 = no reward, 0.5 = reward), distance to the clock game (trials 1 to 6, centered at 0) and their interaction as the independent variables. We hypothesized that there would be a reward by distance interaction, such that memory would be enhanced for trials closer (vs. further) to the clock game in the reward (vs. no reward) condition. We hypothesized that this effect would be present in associative memory, and, like our confirmatory analyses described above, we were agnostic to whether it would be present for item memory. Such distance modulated reward effects would be consistent with the graded effects reported in Braun et al. (2018). We also conducted LMMs with item and associative memory for only the trials immediately before the clock game as our dependent variables and reward (-0.5 = no reward, 0.5 = reward), delay (-0.5 = immediate, 0.5 = delayed), and their interaction as our independent variables. This analysis more closely mirrored the approach used by Murayama & Kitagami (2014). Again, we hypothesized that there would be a delay-dependent effect of reward on associative memory for immediately preceding trials, and, like our confirmatory analyses, we were agnostic to whether such an effect will be present in item recognition for immediately preceding trials.

Fifth and finally, we tested whether reward leads to recognition priming in the item recognition test (Rouhani et al., 2020). We conducted LMMs with corrected recognition in the delayed test as the dependent variable and reward (-0.5 = no reward, 0.5 = reward), priming condition (factor with 3 levels): 1) preceding item from different block, same outcome, 2) preceding item from different block, different outcome, 3) preceding item from same block), and

their interaction as the independent variables. We hypothesized that corrected recognition would be higher on trials in which the preceding item was from the same block items (vs. different block, regardless of outcome) in the reward (vs. no reward) condition.

Pilot Data.

We collected 36 pilot participants on the task described above (see Procedure) to validate our design and provide preliminary evidence for each of our three preregistered hypotheses (see Table 1). Briefly, piloting confirmed that the task produced reasonable levels of performance across participants: There were no exclusions necessary related to task compliance; no observations were at ceiling, and only one participant had below floor memory performance (corrected recognition below 0). Additionally, no participants were excluded for low associative memory performance. Therefore, only one participant was excluded out of 36. Our pilot data further confirmed the effectiveness of our reward manipulation: participants were significantly faster and more accurate on reward vs no reward trials of the clock game (see Supplementary Information). Please see Supplementary Information for a full description and statistical analysis of the pilot data.

Final Sample Characteristics.

At our first two checkpoints (Checkpoint 1: n=90; Checkpoint 2: n=165), we did not find informative evidence for all our hypotheses, such that some preregistered confirmatory analyses neither reached the alpha boundary for a significant effect nor yielded sufficient evidence for the null. As such, we continued data collection until we reached our maximum sample size. We recruited 307 participants and, in accordance with our preregistered exclusion criteria, excluded participants if they received fewer than 6 rewards during the clock game (n = 13), if they failed to respond during the flanker task (n = 1), and if they performed below chance on the immediate

or delayed memory tests, irrespective of reward (n = 50). Our final sample contained 244 participants (mean age = 28.61 years, 125 female, mean years of education = 15.35).

Transparency and Openness.

All pilot data are publicly available on Open Science Framework (https://osf.io/35fcs/). Behavioural data from the full experiment will be anonymized and made publicly available upon acceptance of the manuscript for publication. All randomization, experiment, and analysis scripts, along with resulting simulations from the pilot data are publicly available on Open Science Framework (https://osf.io/35fcs/). Analysis scripts from the full experiment are also publicly available. All pilot analyses were conducted using R version 3.6.1. The study design and analysis approach were registered and went through peer review prior to data collection.

Results

Preregistered Confirmatory Analyses

We first conducted analyses to confirm that participants performed above chance on both the item (chance = 0) and associative memory (chance = 0.333) tests. Participants successfully recognized items in both the immediate and delayed item memory tests (Immediate: mean corrected recognition = 0.663, $t(243) = 70.126$, $p < 0.0001$, Delayed: mean corrected recognition = 0.329, $t(243) = 37.465$, $p < 0.0001$) and also remembered associations above chance in the immediate and delayed associative memory tests (Immediate: mean = 73.32%, $t(243) = 66.45$, $p < 0.0001$; Delayed: mean = 53.48%, $t(243) = 61.07$, $p < 0.0001$).

Turning to our main questions of interest, we next assessed whether there was a delay-dependent effect of reward on associative memory (Hypothesis 1a; Figure 2a). Contrary to our prediction, there was no significant main effect of reward (beta = 0.004, 95% CI = [-0.011,

0.019], p = 0.607). Although associative memory performance declined from the immediate to the delayed test (beta = -0.198, 95% CI = [-0.214, -0.183], p < 0.0001), there was no significant interaction between reward and delay (beta = 0.018, 95% CI = [-0.013, 0.048], p = 0.250). Having failed to detect a delay-dependent reward effect using frequentist analyses, we turned to our preregistered Bayesian analyses. These analyses provided evidence for a null effect of reward (% inside ROPE: 100%; see *Supplementary Information* for all Bayesian model outputs) but not definitive evidence for a null interaction between reward and delay (% inside ROPE: 85.99%). Thus, we did not find support for our first preregistered hypothesis of a delay-dependent effect of reward on associative memory.

We next examined the effect of reward on associative memory separately in the immediate and delayed tests. Reward did not influence associative memory in the immediate test (Hypothesis 1b, beta = 0.005, 95% CI = [-0.023, 0.033], p = 0.969, % inside ROPE: 100%). In contrast to Hypothesis 1c, reward also did not influence associative memory in the delayed test (beta = -0.013, 95% CI = [-0.041, 0.015], p = 0.641, % inside ROPE: 99.3%). Together, these results suggest that reward did not retroactively enhance associative memory either immediately or after a delay.

We next assessed whether a delay-dependent effect of reward emerged in item memory (Hypothesis 2a; Figure 2b). There was no main effect of reward on item memory (beta = -0.004, 95% CI = [-0.018, 0.009], p = 0.543, % inside ROPE: 100%)) and no reward-by-delay interaction (beta = 0.012, 95% CI = [-0.015, 0.040], p = 0.381, % inside ROPE: 98%). Similarly, we found no effect of reward on item memory when examined separately in the immediate test (Hypothesis 2b; beta = 0.010, 95% CI = [-0.015, 0.036], p = 0.720, % inside ROPE = 100%) or the delayed test (Hypothesis 2c; beta = -0.002, 95% CI = [-0.028, 0.024], p = 0.997, % inside ROPE = 100%). These findings indicate that reward did not influence item memory

performance at either delay.

Finally, we tested whether the effect of reward differed across associative and item memory in the delayed test (Hypothesis 3; Figure 2c). There was no main effect of reward (beta = 0.037, 95% CI = [-0.027, 0.100], p = 0.264, % inside ROPE: 100%) nor an interaction between reward and memory test on standardized memory performance (beta = 0.058, 95% CI = [-0.069, 0.185], p = 0.374, % inside ROPE: 96.68%). In a preregistered exploratory analysis (see Methods), there was inconclusive evidence for or against a three-way interaction among reward, delay, and memory test (beta = 0.042, 95% CI = [-0.156, 0.239], p = 0.679, % inside ROPE = 91.50%).

In summary, we found no evidence for a delay-dependent retroactive effect of reward on either item or associative memory. Instead, our Bayesian analyses generally supported the null hypothesis. As described in our Analysis Plan, we also repeated our preregistered, confirmatory analyses on the subset of participants who performed statistically above chance on all measures. However, even in these high performing participants, we did not find any significant effects of reward on memory (see *Supplementary Information*). These findings diverge from studies reporting retroactive reward effects (Murayama & Kitagami, 2014; Patil et al., 2017; Braun et al., 2018) on item recognition but are consistent with other work showing null effects (Oyarzún et al., 2016; Sukumaran et al., 2023), prompting further consideration of potential methodological or contextual factors underlying these discrepant findings.

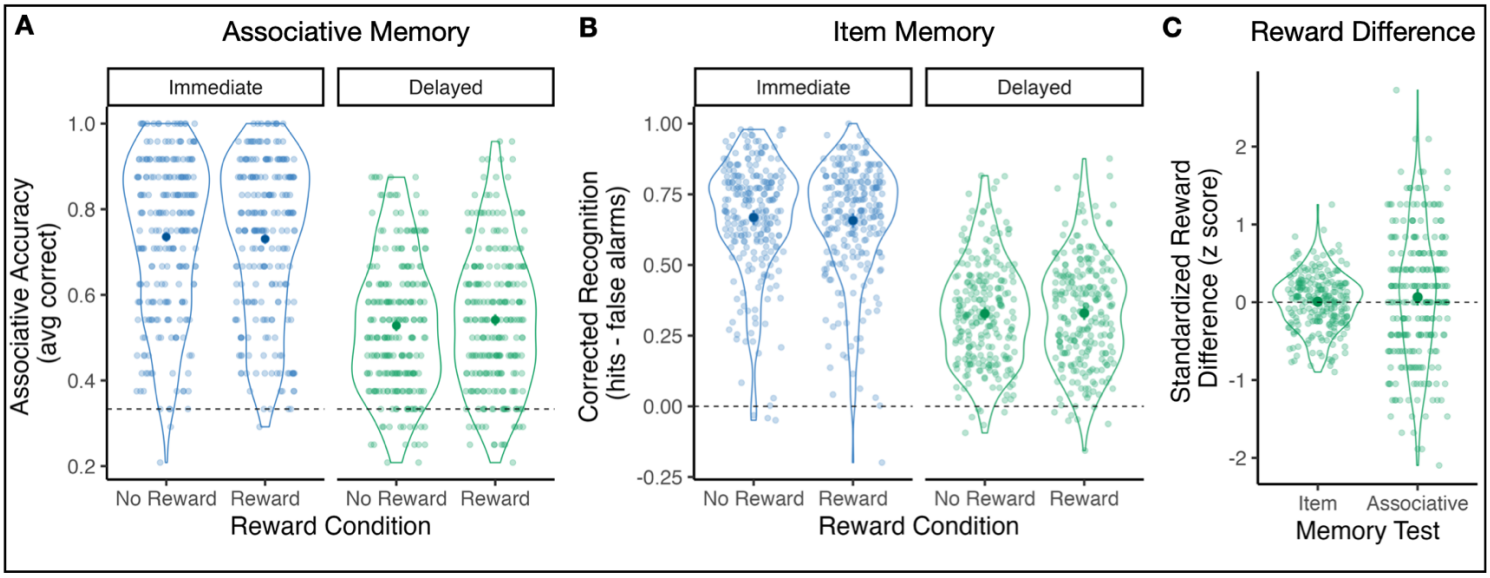


Figure 2 | Memory performance as a function of reward and delay. (A) Associative memory performance was not influenced by reward, delay, or their interaction (Hypothesis 1). (B) Item memory performance was likewise unaffected by reward, delay, or their interaction (Hypothesis 2). (C) Reward effects did not differ between associative and item memory tests at the long delay (Hypothesis 3). Large points represent group means; small points represent individual participants. Error bars denote 95% confidence intervals. The dashed line indicates chance performance. Blue indicates the immediate test and green indicates the delayed test.

Preregistered Exploratory Analyses

We next conducted several preregistered exploratory analyses to investigate additional factors that might reveal a retroactive effect of reward on memory. Specifically, we assessed (1) item memory for high confident judgements; (2) associative memory on trials for which both constituent items were correctly recognized; (3) whether proximity to reward influences memory by both (a) characterizing the continuous effect of distance to reward and (b) restricting analysis to trials encoded immediately preceding reward; and (4) whether hit rate was higher for trials tested following items that were encoded in the same block (see *Methods* for details).

First, we repeated our confirmatory analyses on item memory but included only high-confidence responses, following the approach taken in prior work (Patil et al., 2017). However, there was no effect of reward on the accuracy of high confidence item memory (beta = -0.003, 95% CI = [-0.02, 0.015], p = 0.713, % in ROPE = 100%) and no reward-by-delay interaction

(beta = 0.009, 95% CI = [-0.028, 0.046], p = 0.620, % in ROPE = 98.34%). Planned contrasts similarly revealed no significant reward effect in either the immediate (beta = 0.008, 95% CI = [-0.026, 0.042], p = 0.928, % in ROPE = 95.00%) or delayed test (beta = -0.001, 95% CI = [-0.036, 0.033], p = 0.997, % in ROPE = 100%). Thus, reward did not enhance memory for confidently recognized items.

Second, we may not have been able to observe an effect of reward on associative memory because participants may not have recognized the underlying items forming each association. To examine this, we analyzed only associative memory trials in which participants correctly recognized both constituent items. Even under this stricter criterion, we found no main effect of reward (beta = 0.018, 95% CI = [-0.007, 0.043], p = 0.171, % in ROPE = 99.14%) and no conclusive evidence for or against a reward-by-delay interaction (beta = 0.021, 95% CI = [-0.029, 0.072], p = 0.416, % in ROPE = 80.50%). Once again, planned contrasts failed to reveal an effect of reward in the immediate test (beta = -0.007, 95% CI = [-0.053, 0.039], p = 0.978, % in ROPE = 100%) and in the delayed test (beta = -0.028, 95% CI = [-0.077, 0.019], p = 0.428, % in ROPE = 95.29%), suggesting that the absence of a retroactive reward effect on associative memory was not due to poor item memory.

We next considered whether the retroactive influence of reward might depend on temporal proximity to the reward. Specifically, we tested whether there was a graded effect of reward on memory, such that effects were strongest for items encoded immediately before a reward and weaker for those further away (Braun et al., 2017). However, we found no evidence for an interaction between distance to reward and reward for either associative memory (beta = 0.001, 95% CI = [-0.009, 0.012], p = 0.846, % in ROPE = 100%) or item memory (beta = 0.003, 95% CI = [-0.003, 0.010], p = 0.290, % in ROPE = 100%). Paralleling the approach of Murayama & Kitagami (2014), we next restricted our analysis to only the trials immediately

preceding the reward (i.e., the "pre-reward" trials). We asked whether memory for these trials was modulated by reward, delay, or their interaction. In the Associative Memory Test, we found no effect of reward (beta = 0.025, 95% CI = [-0.005, 0.055], p = 0.101, % inside ROPE = 95.76%) and no conclusive evidence for or against a reward-by-delay interaction (beta = 0.011, 95% CI = [-0.049, 0.071], p = 0.713, % inside ROPE = 90.63%, Figure 3a) for these pre-reward trials. In contrast, in the Item Memory Test, we observed a main effect of reward, such that items directly preceding a reward were remembered better than those in the no-reward condition (beta = 0.027, 95% CI = [0.004, 0.050], p = 0.018; Figure 3b), but we did not find conclusive evidence for a reward by delay interaction (beta = 0.026, 95% CI = [-0.019, 0.071], p = 0.260, % inside ROPE = 91.58%). Follow-up contrasts revealed that there was no main effect of reward in the immediate test (beta = -0.0143, 95% CI = [-0.057, 0.028], p = 0.818), but a trending effect of reward after a delay (beta = -0.04, 95% CI = [-0.083, 0.002], p = 0.065). This finding is similar to Murayama and Kitagami (2014), who also reported enhanced memory for items that immediately preceded a reward after a delay. Nevertheless, it is important to note that this analysis was based on a limited number of trials, which may reduce the reliability of this effect.

In our final preregistered exploratory analysis, we tested whether reward influenced recognition priming in the delayed item memory test. Specifically, we examined whether hit rate in the item memory test was affected by reward, the block type of the preceding test item (same encoding block, different block with the same outcome, or different block with a different outcome), and their interaction. We found no evidence of recognition priming. Instead, hit rate was lower for items preceded by same-block items compared to those preceded by items encoded in a different block with a different outcome (beta = -0.034, 95% CI = [-0.05, -0.016], p = 0.0002), and this difference did not interact with reward (beta = -0.031, 95% CI = [-0.005, 0.068], p = 0.096). Hit rates did not differ between the two different-block conditions (same vs.

different outcome; beta = -0.011, 95% CI = [-0.029, 0.007], p = 0.227), and there was no interaction with reward (beta = 0.031, 95% CI = [-0.005, 0.068], p = 0.095). However, the lack of recognition priming overall makes it difficult to interpret the lack of reward-related modulation.

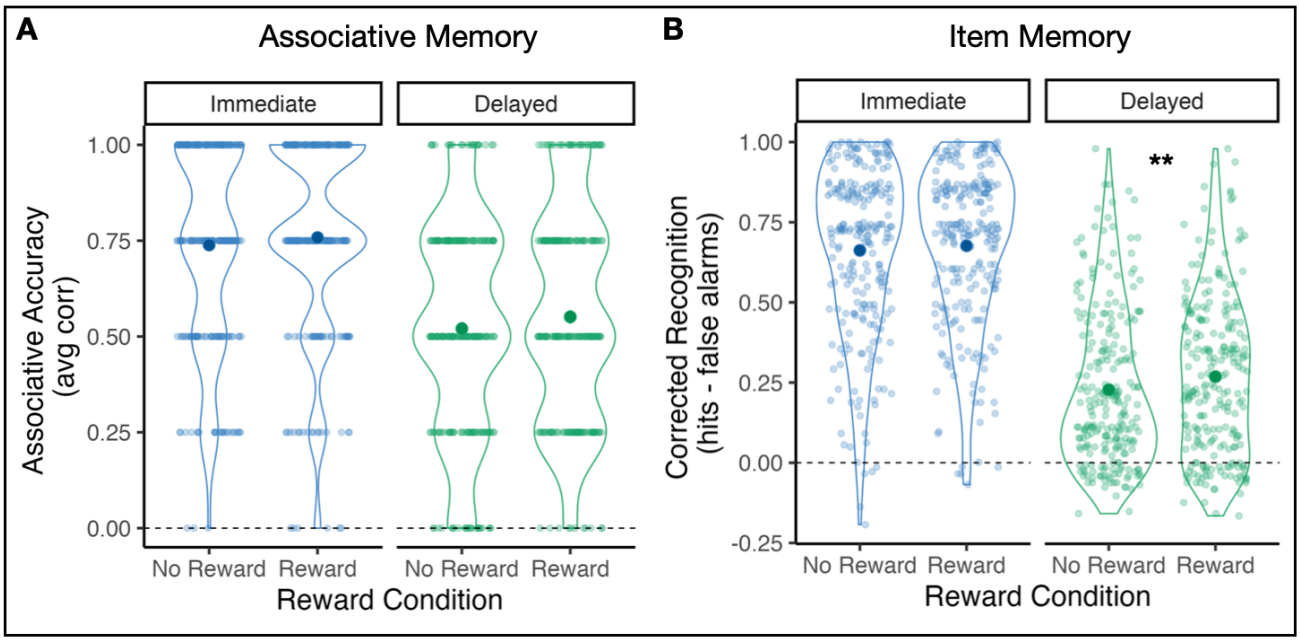


Figure 3 | Memory performance for trials immediately preceding the reward. (A) Associative memory for immediately preceding trials was not influenced by reward, delay, or their interaction. (B) Item memory for immediately preceding trials was higher in the reward vs. no reward condition, but there was no interaction with delay. Large points represent group means; small points represent individual participants. Error bars denote 95% confidence intervals. The dashed line indicates chance performance. Blue indicates the immediate test and green indicates the delayed test.

Exploratory Analyses

Across multiple analyses, we examined several potential ways in which reward might have influenced memory, but generally failed to find evidence for significant retroactive effects on both associative and item memory. One possible explanation for the absence of reward modulation is that our reward manipulation may not have been effective for all participants—that is, if some participants were insensitive to the reward, any reward-related effects on memory would be unlikely to emerge when considering the group as a whole. To address this possibility, we conducted an exploratory analysis examining whether individual differences in

reward sensitivity were related to retroactive reward effects on memory.

We quantified reward sensitivity as the difference in response time during the clock game between non-rewarded and rewarded trials. Higher values indicate faster responses on rewarded relative to non-rewarded trials, reflecting greater reward speeding (Figure 4a). We then calculated the Spearman's correlation between 1) each participant's reward speeding score and 2) their difference in associative memory performance between the rewarded and non-rewarded conditions at the long delay. This analysis revealed a positive relationship: participants who exhibited greater reward speeding—reflecting stronger sensitivity to the reward manipulation—also showed larger reward-related enhancements in associative memory at the long delay (rho = 0.193, p = 0.002; Figure 4b). There was a trending relationship between reward speeding and associative memory in the immediate test, but in the opposite direction, such that weaker reward sensitivity was associated with better memory (rho = -0.125, p = 0.052). Indeed, there was a significant interaction between reward speeding and delay on associative memory (beta = -0.297, 95% CI = [-0.47, -0.121], p = 0.0009). There was a trending relationship between reward speeding and item memory in the immediate test (Immediate: rho = 0.119, p = 0.063), but not the delayed test (Delayed: rho = -0.059, p = 0.359). Again, there was a trending interaction between reward speeding and delay on item memory, but this effect was in the opposite direction, such that reward speeding positively predicted item memory in the immediate, rather than the delayed, test (beta = 0.176, 95% CI = [-0.0073, 0.354], p = 0.052). Together, these analyses suggest that reward speeding was correlated with memory performance, raising the possibility that individual differences in reward sensitivity may contribute to the relationship between reward and memory.

To examine this possibility, we next repeated our confirmatory analyses in the subset of participants who demonstrated sensitivity to the reward manipulation. We defined reward-

sensitive participants as those whose reward speeding score was above the median (36.25 ms; Figure 4c). In this subset, we found evidence for our initial hypotheses: there was an interaction between reward and delay on associative memory (beta = 0.076, 95% CI = [0.034, 0.117], p = 0.0003; Figure 4d). Follow-up contrasts indicated inconclusive evidence for an effect of reward on associative memory in the immediate test (beta = 0.029, 95% CI = [-0.009, 0.067], p = 0.215, % in ROPE = 64.23%), but a significant effect in the delayed test (beta = -0.047, 95% CI = [-0.085, -0.008.], p = 0.010).

Although reward-sensitive participants showed a delay-dependent effect of reward on associative memory, we still failed to find a reward-by-delay interaction in the item memory test (beta = 0.006, 95% CI = [-0.035, 0.047], p = 0.775, % in ROPE = 98.68%, Figure 4e). There was no effect of reward on item memory in either the immediate (beta = 0.007, 95% CI = [-0.031, 0.046], p = 0.956, % in ROPE = 98.87%) or the delayed test (beta = 0.002, 95% CI = [-0.037, 0.039], p = 0.999, % in ROPE = 100%). Thus, the delay-dependent effect of reward emerged only for associative memory, not for item memory, among reward-sensitive participants.

Finally, we tested whether the effect of reward on associative memory was greater than that on item memory (Figure 4f). In the delayed test, there was a significant main effect of reward (beta = 0.115, 95% CI = [0.030, 0.199], p = 0.008) and an interaction between reward and memory test (beta = 0.242, 95% CI = [0.073, 0.411], p = 0.005), with greater reward modulation of associative vs. item memory. Moreover, we observed a three-way interaction between reward, delay, and memory test (beta = 0.358, 95% CI = [0.079, 0.636], p= 0.012), such that there was a delay-dependent effect of reward on associative—but not item—memory. Together, this provides support for each of our hypotheses: reward retroactively enhances associative, but not item, memory in a delay dependent manner. Further, these results also

highlight an intuitive but overlooked prerequisite for this effect: individuals must be sensitive to the reward for it to influence memory.

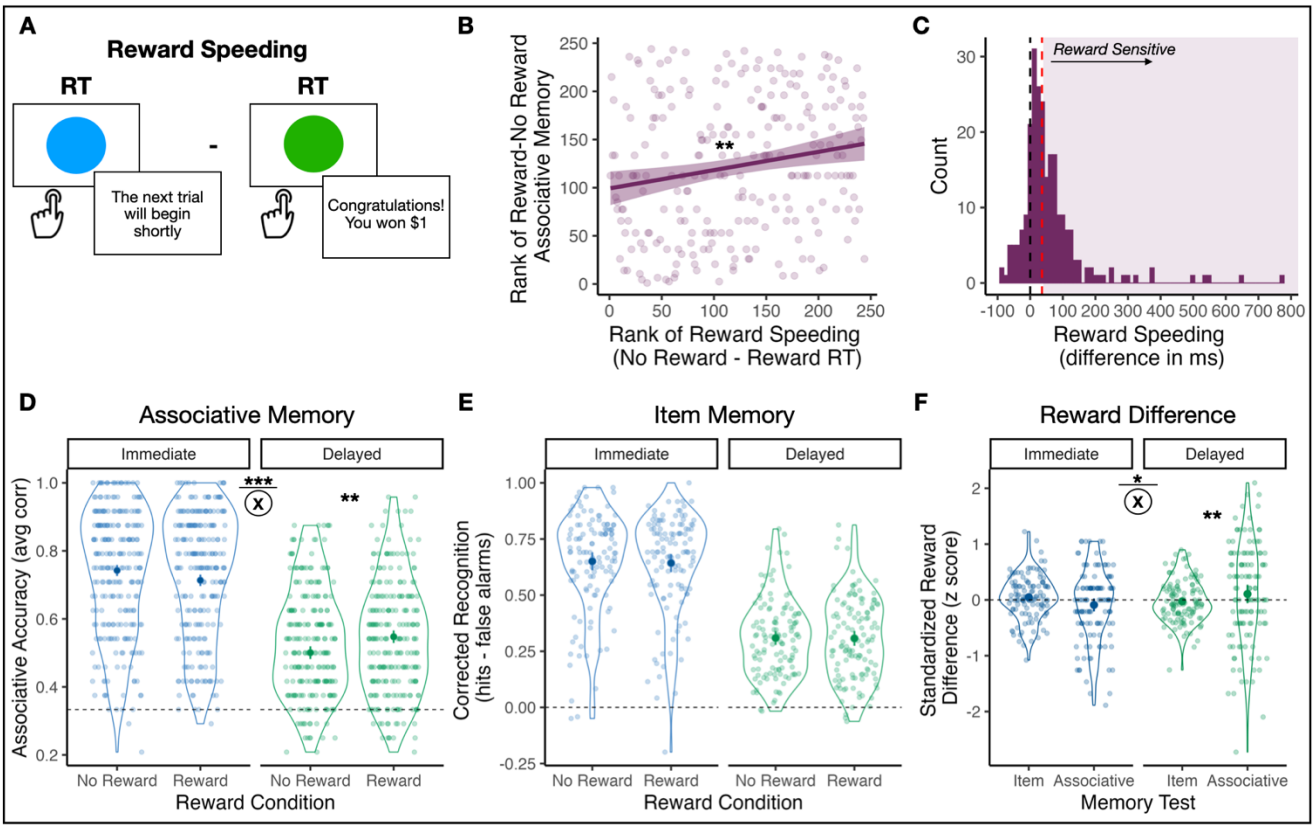


Figure 4 | Memory performance in reward-sensitive participants. (A) Reward speeding was calculated as the difference in response times between non-rewarded and rewarded trials in the clock game. (B) Reward speeding was positively correlated with the difference in associative memory between reward conditions at the long delay. Purple dots represent individual participants; the purple line shows the Spearman correlation, and the shaded ribbon indicates the 95% confidence interval. (C) Distribution of reward speeding scores across all participants. The black line marks a reward speeding score of 0, and the red line marks the group median. Participants with reward speeding scores above the median were classified as reward-sensitive and included in subsequent analyses. (D) In reward sensitive participants, there was a significant interaction between reward and delay on associative memory. Associative memory was better in the reward vs. no reward condition, but only at the long delay. (E) In reward sensitive participants, there was no effect of reward, delay, or their interaction. (F) In reward sensitive participants, the delay-dependent effect of reward was greater for associative, compared to item, memory. Large points represent group means; small points represent individual participants. Error bars denote 95% confidence intervals. The dashed line indicates chance performance. Blue indicates the immediate test and green indicates the delayed test.

Discussion

In this Registered Report, we investigated whether reward retroactively enhances the

consolidation of associative and item memory, drawing on neurobiologically grounded theories of the dopaminergic modulation of the hippocampus to address discrepant findings in the human literature. Our planned confirmatory analyses did not reveal any evidence for retroactive effects of reward on memory. However, in a set of exploratory analyses we found evidence for all three of our initial hypotheses but, critically, these effects emerged only in the subset of participants who were behaviourally sensitive to the reward manipulation. Among these individuals, reward enhanced associative but not item memory, and did so only after a delay. Moreover, this delay-dependent retroactive enhancement was significantly larger for associative than item memory. Together, these findings align with long-standing neurobiological accounts of how dopamine shapes hippocampal consolidation and delineate key boundary conditions on retroactive reward effects, offering a unifying explanation for inconsistencies in past studies.

We found that reward sensitivity was a central factor in determining whether retroactive effects on memory emerged. Unplanned exploratory analyses showed that participants who were more behaviourally responsive to reward, indexed by reaction-time modulation during the clock game, did show retroactive enhancements. This highlights an important boundary condition of this effect: retroactive reward benefits appear only when participants are sufficiently motivated by the reward. Mechanistically, reward-sensitive individuals may have experienced greater dopamine release, producing stronger modulation of hippocampal-dependent memories over time. This interpretation aligns with work linking individual differences in reward responsiveness to activity in the ventral striatum, an important target of dopaminergic midbrain projections (Kim et al., 2015). Past studies reporting positive retroactive reward effects (Murayama & Kitagami, 2014; Patil et al., 2017; Braun et al., 2018) may simply have included more reward-responsive participants or offered more motivating rewards compared to those with null findings (Oyarzún et al., 2016; Sukumaran et al., 2023). Indeed, our pilot data, which were

collected during the peak of the COVID-19 pandemic, showed evidence for retroactive reward effects at the group level (see *Supplementary Information*). One speculative possibility for why we observed stronger overall reward modulation in this smaller sample (n = 36) is because the rewards may have been more motivationally salient during this period. Additionally, smaller samples are more likely to yield uneven representations of reward sensitivity, potentially biasing reward effects. By contrast, the larger sample in the present study (n = 244) enabled us to detect variability in reward responsiveness across individuals and relate it to memory performance. Together, these findings highlight the importance of incorporating direct measures of reward sensitivity and carefully considering reward salience when designing future retroactive reward paradigms to clarify when and for whom such effects emerge.

Among reward-sensitive participants, reward selectively enhanced associative memory, and only after a delay. This pattern aligns with long-standing neurobiological models in which dopamine modulates hippocampal consolidation over time (Bethus et al., 2010; Lisman et al., 2011; O'Carroll et al., 2006; Rossato et al., 2009), given that associative memories are known to rely on the hippocampus (Eichenbaum, 2000; Davachi, 2006). In contrast, we found limited evidence for retroactive reward effects on item memory. In our confirmatory analyses when including all six trials preceding the reward, there was no evidence for reward modulation of item memory, even when examining only reward-sensitive individuals. However, when focusing on the single trial immediately preceding the reward, we did observe a retroactive effect on memory. Follow up planned contrasts within each delay suggested this effect showed a trend only at a delay, and was not significant at the immediate test. This finding parallels the results of Murayama and Kitagami (2014) but raises the question of why the retroactive effect on item memory was different and less robust than that observed for associative memory. As described in the Introduction, one possibility is that item recognition is less likely to rely on the

hippocampus (Brown & Aggleton, 2001; Davachi et al., 2003; Davachi & Wagner, 2002; Eldridge et al., 2000; Wais, et al., 2011; Wan et al., 1999), suggesting a different route through which reward may modulate item versus associative memories. Another not mutually exclusive possibility is that the selective enhancement of immediately preceding trials reflects a more limited temporal reach of reward on item memory, consistent with prior work showing stronger modulation for items closer to, rather than further from, the reward (Braun et al., 2018). Although, it is important to note that we did not observe such temporally graded effects overall in our data.

Further, other design differences between our study and previous retroactive reward work may have contributed to differences in overall item memory effects. For example, Patil et al. (2017) found retroactive effects in high-confidence responses. This trial restriction may have allowed them to focus on hippocampal dependent memories (Rugg et al., 2012). However, even high-confidence item trials in our study showed no retroactive effect, suggesting this explanation may be insufficient on its own. A second difference concerns how neutral items were linked to subsequent reward. Patil et al. (2017) examined retroactive tagging at the category level, such that rewarding a category retroactively enhanced memory for earlier neutral items from that category. In this case, items and rewards were connected through a shared semantic structure, which may provide a more stable scaffold for reward signals to propagate backward in time (Dunsmoor et al., 2022). In contrast, in our study, reward was linked to preceding items solely through temporal proximity, without an explicit semantic connection. One possibility, therefore, is that categorical relationships support broader or more robust reward-related modulation, potentially via reactivation of semantically organized representations, whereas purely temporal associations yield more limited retroactive effects on item memory. Although, note that other work using a similar category structure has failed to find retroactive items effects for both

reward- (Sukumaran et al., 2013) and fear-based tagging (Kalbe & Schwabe, 2022).

By focusing on hippocampally dependent associative memories, our study helps clarify when retroactive reward effects emerge and provides a mechanistic framework that may reconcile mixed findings in the literature. One plausible mechanism underlying these results is the synaptic tag-and-capture hypothesis, in which dopamine release following reward receipt selectively strengthens recently active synapses in the hippocampus. According to this framework (Frey & Morris, 1997, 1998), synapses activated during an earlier experience can make use of plasticity-related proteins released during subsequent dopamine bursts. Such plasticity-related proteins enhance late long-term potentiation, allowing synapses that code for memories of preceding experiences to be preferentially consolidated (de Carvalho Myskiw et al., 2013; Dunsmoor et al., 2022; Moncada & Viola, 2007; Morris, 2006; Salvetti et al., 2014; Wang et al., 2010). In our study, dopamine released by reward may have allowed hippocampal representations of preceding items to benefit from dopamine-dependent consolidation, leading to the emergence of reward effects on associative memory only after a delay. This mechanism provides a biologically grounded explanation for the delay dependence of the effect, its specificity to hippocampally-dependent associative memories, and its reliance on reward sensitivity.

These findings have important implications for theories of reward learning and adaptive memory modulation by shedding light on how our memory systems prioritize experiences whose relevance only becomes apparent after the fact. Such prioritization is central to the temporal credit assignment problem, where we must link outcomes to earlier events to form meaningful predictions and guide future behavior (Brzosko et al., 2015; Hull, 1943; Izhikevich, 2007). Importantly, our results show that retroactive reward selectively enhances associative, rather than item, memory, and that this effect is delay-dependent**,** suggesting associative

relationships rather than isolated items may be prioritized for the long haul. This distinction has implications for decision making: Associative memories may provide richer, more flexible information for linking past events to outcomes, enabling more accurate predictions and adaptive choices in complex environments (Biderman et al., 2020; Wimmer & Shohamy, 2012). At the same time, the effect we observed was not particularly robust and was only present in a subset of reward-sensitive participants, which suggests that retroactive reward enhancement alone is unlikely to fully account for temporal credit assignment in real-world learning. Instead, these findings may indicate one mechanism by which relevant memories are prioritized, but opens additional questions about other factors that might interact with retroactive memory modulation to guide adaptive behaviour. For example, how do factors such as goal relevance (Mather & Sutherland, 2011; Cowan et al., 2021) or uncertainty (Nicholas et al., 2022) influence the likelihood that a particular experience will be prioritized? Continued exploration of these questions will help clarify how retroactive reward effects operate alongside other learning processes to shape adaptive decision making.

Our findings also have important implications for understanding when reward-based memory modulation may succeed or fail across different populations, such as in Parkinson's disease or during healthy aging. For example, Parkinson's patients show improved memory for neutral information associated with rewards when on, compared to off, dopaminergic medication (Sharp et al., 2020). Extending these findings to examine the retroactive modulation of memories could further inform our understanding of reward-related memory changes in Parkinson's disease. In contrast, older adults' memory for rewarded information often remains relatively intact (Mather & Schoeke, 2011; Spaniol et al., 2014; Swirsky et al., 2021) despite variable age-related declines in dopamine (Bäckman et al., 2000; Karalija et al., 2022). Notably, these studies generally used monetary rewards to motivate encoding. However, other work

suggests that older adults show shifts in motivational preferences, placing relatively greater value on socioemotional vs. monetary rewards compared with younger adults (Rademacher et al., 2013; Swirsky et al., 2023). These findings imply that while reward can modulate memory across the lifespan, the type of reward matters: monetary incentives may be less effective for some older adults, whereas socially meaningful rewards could potentially exert stronger memory-enhancing effects (Swirsky et al., 2023). Our results underscore the importance of taking this variability in reward sensitivity into account, opening the possibility of designing targeted interventions to enhance memory by leveraging individuals' motivational preferences.

Together, our study provides behavioral evidence that retroactive reward selectively enhances associative, but not item, memories in a delay-dependent manner, but only in individuals who are sensitive to reward. These findings highlight hippocampally dependent associative memories and individual differences in reward sensitivity as key boundary conditions for retroactive reward effects, offering a mechanistic account that helps reconcile prior inconsistencies in the literature. More broadly, our results advance our understanding of the adaptive nature of memory, showing how positive experiences can reach back in time to strengthen the formation of long-lasting associative memories that can be used to guide future behavior.

Constraints on Generality

We used several inclusion criteria with the goal of maintaining experiment validity and matching critical aspects of the populations studied in the inspiring work (Patil et al., 2017; Braun et al., 2018; Murayama & Kitagami, 2014). Like the prior work, we restricted our sample to young adults (18-30 years of age). This restriction limits the generalization of our results, but is well motivated because the dopaminergic system and sensitivity to reward can vary across the life span (Dreher et al., 2008; Wahlstrom et al., 2010). Thus, if we failed to replicate the

influence of reward on memory in an older sample, the age of the sample could be a confounding variable. Future work will be needed to expand the generalizability of our results to other ages, including but not limited to adolescents and older adults. We also required that participants self-report being proficient in English so that they could understand the task instructions, and having normal or corrected to normal vision so that they could clearly perceive the stimuli. These criteria also limit the generalization of our findings but were required for the implementation of the experiment.

With these constraints, we aimed to recruit an otherwise diverse sample using CloudResearch. One notable limitation of CloudResearch is that approximately 95% of active MTurk workers (from which CloudResearch recruits) are based in the United States. As such, we limited our sample to those living in the United States. Additionally, demographics distributions from CloudResearch suggest that 75 to 80% of their workers identify as White. Indeed, over 50% of our final sample identified as White (n = 152), with the remaining participants identifying as Mixed Race, Black, Native American, Hispanic, Asian, and Middle Eastern. Prior work has found retroactive reward effects in samples from the United States (Patil et al., 2017; Braun et al, 2018) and Japan (Murayama & Kitagami, 2014), suggesting that this effect may generalize across countries and cultures. However, future work will be needed to determine the generalizability of our findings to more globally diverse samples, including individuals living in countries other than the United States, and to determine how motivational preferences vary across cultural contexts (Blevins et al., 2023). Such work will be important for understanding how reward-related memory enhancements may differ as a function of country and culture.

Finally, because sensitivity to monetary reward may differ as a function of socioeconomic status (White et al., 2022), we also collected information on participants' socioeconomic status. However, we did not find a relationshipsbetween reward modulation of

memory and socioeconomic status (see *Supplementary Information*). However, CloudResearch workers in the US tend to belong to somewhat higher socioeconomic groups than the general US population so our effects reflect this sampling. Future work will benefit from systematically testing the effect of reward on individuals from varying socioeconomic backgrounds.

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Author contributions

HTS, MLS, and KDD conceptualized the research question, experiment design, and analytic approach. HTS collected the data and performed the analysis. MLK and KDD acquired funding and supervised. HTS wrote the original draft and MLK and KDD revised and edited the draft.